

RemedyFabric

Autonomous repair that remains safe when a repair Agent crashes, hallucinates, equivocates, replays evidence, or crosses its role boundary.

No single Worker can authorize release.

PRE-FREEZE EVIDENCE SET

EVIDENCE COMPLETE

AgentTeams official runtime:
validated
Clean wheel replay: isolated
package replay passed

Apache-2.0 / public repository

A repair Agent can become the incident

01**Plausible output**

A convincing patch can be unsafe, stale, or scoped beyond permission.

02**Correlated checks**

More votes do not help when every role trusts the same evidence.

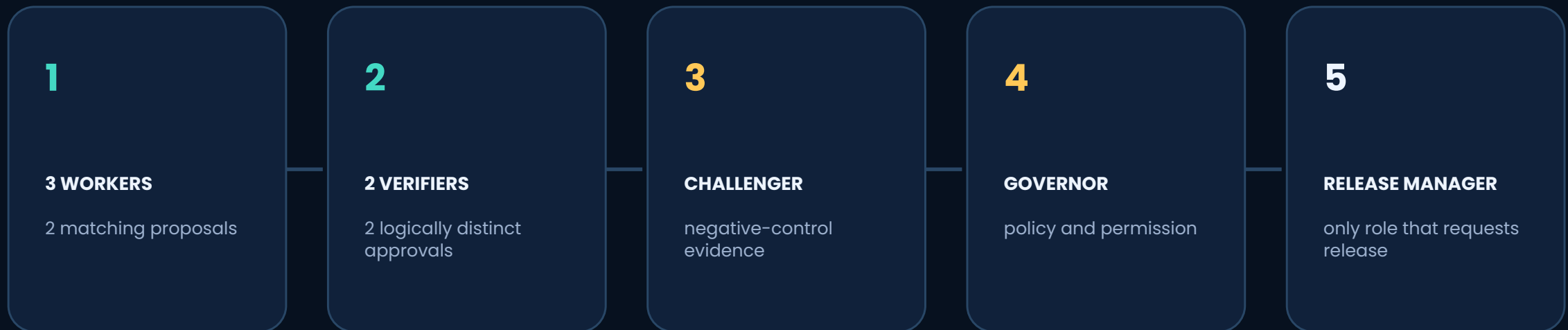
03**Unclear terminal state**

A timeout must end in a verified release or byte-exact restore.

WINNING CONTRACT

Within the declared one-Worker Byzantine-like boundary: recover without unsafe release. Beyond it: refuse release and restore. Every decision binds run, snapshot, candidate and evidence digests.

Role-separated identities and release authority



Default release equation: 2 Worker proposals + 2 Verifiers + 1 Challenger + 1 Governor + registered Release Manager. Duplicate identities do not inflate quorum; rejection, equivocation, stale context, forged or replayed evidence selects rollback. The local matrix uses logically distinct identities and isolated clones; shared implementation or provider failures are outside its one-Worker claim.

Recovery survives one faulty Worker in the tested matrix

ONE-FAULT RECOVERY

100%

8 executable trials

UNSAFE RELEASES

0

within one-Worker tested boundary

OVERFLOW FAIL CLOSED

100%

13 enumerated trials

EXPLORED STATES

20,748

finite exhaustive model

INJECTED CLASSES

crash / timeout / unsafe patch / equivocation / forged proposal or evidence / replay / context mismatch / role impersonation

CLAIM BOUNDARY

Finite one-Worker Byzantine-like model with trusted membership and role-separated logical controls. Correlated implementation/provider failure is out of scope. Not general BFT, cryptographic identity, host security, or production safety proof.

Real provenance; explicit protocol-simulation boundary

CASES

30

merged OSS PR identities

REPOSITORIES

6

official GitHub REST verified

LANGUAGES

3

Python / JavaScript / Rust

FAULT CLASSES

8

all case transformations named

CHAMPION PROTOCOL RESULT

23 within-boundary cases; safe recovery 100%

Unsafe release 0%

7 overflow/control-plane cases; containment 100%

WHAT WAS ACTUALLY EXECUTED

3 independent extracted semantics passed. The 30 upstream repos were not cloned, repaired, or tested end-to-end; these are typed-message evaluations, not 30 autonomous repair successes.

Every green claim has a machine-readable receipt

VERIFIED

GitHub provenance

30/30 cases; source receipt is evidence-bound

VERIFIED

Micro-replays

3 fixtures, 3 languages

VERIFIED

Container isolation

network none, read-only rootfs, no capabilities, bounded resources

VERIFIED

Clean wheel replay

isolated package replay passed; not evidence for a later package

VERIFIED

Official AgentTeams

validated live receipt; 8 Worker containers

The local container probe is one execution, not a claim about every platform. Public CI, official cloud API calls, production use and external reliability are separate facts.

The standout mechanism maps to every scoring dimension

25%**Scenario value**

Repair infrastructure remains safe when its own Agents fail.

25%**Multi-Agent loop**

Workers, Verifiers, Challenger, Governor and Release Manager.

25%**Skill and ecosystem**

Typed PatchCandidate/Attestation contracts; reusable recovery Skill.

20%**Engineering and safety**

Quorum, rollback, isolation, receipts, model exploration, negative controls.

5%**Open contribution**

Apache-2.0 code, benchmark corpus, fixtures, docs and evidence dashboard.

Coverage mapping is not a predicted score or award guarantee.

Clone. Run. Inspect the receipts.

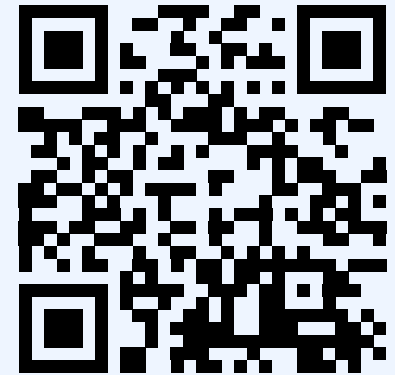
```
$ python3 -m pip install -e .
```

```
$ remedyfabric resilient --output artifacts/resilient-matrix.json
```

```
$ python3 scripts/run_faultbench.py
```

```
$ python3 scripts/run_faultbench_micro_replays.py
```

```
$ python3 scripts/champion_gate.py
```



Repository and evidence

github.com/Oxygen56/remedyfabric

Apache-2.0 / Python 3.11+ / deterministic zero-credential path

What a judge may safely conclude

SUPPORTED

- 22 executable fault-matrix trials
- 20,748 finite model decisions
- 30 public provenance records verified
- 3 extracted semantics executed
- one Docker isolation probe
- explicit cost/provider boundaries

NOT CLAIMED

- general Byzantine consensus
- cryptographic workload identity
- 30 autonomous OSS repairs
- production adoption or reliability
- third-party certification
- external SOTA or guaranteed rank
- cloud API use unless a live receipt says so

MACHINE-READABLE EVIDENCE

[artifacts/champion-evidence.json](#)